**Student Marks Database**

**End-to-End MySQL Project**

*ER Diagram • Schema • Sample Data • Indexes • Views • Stored Procedures • Triggers • Query Bank with Answers*

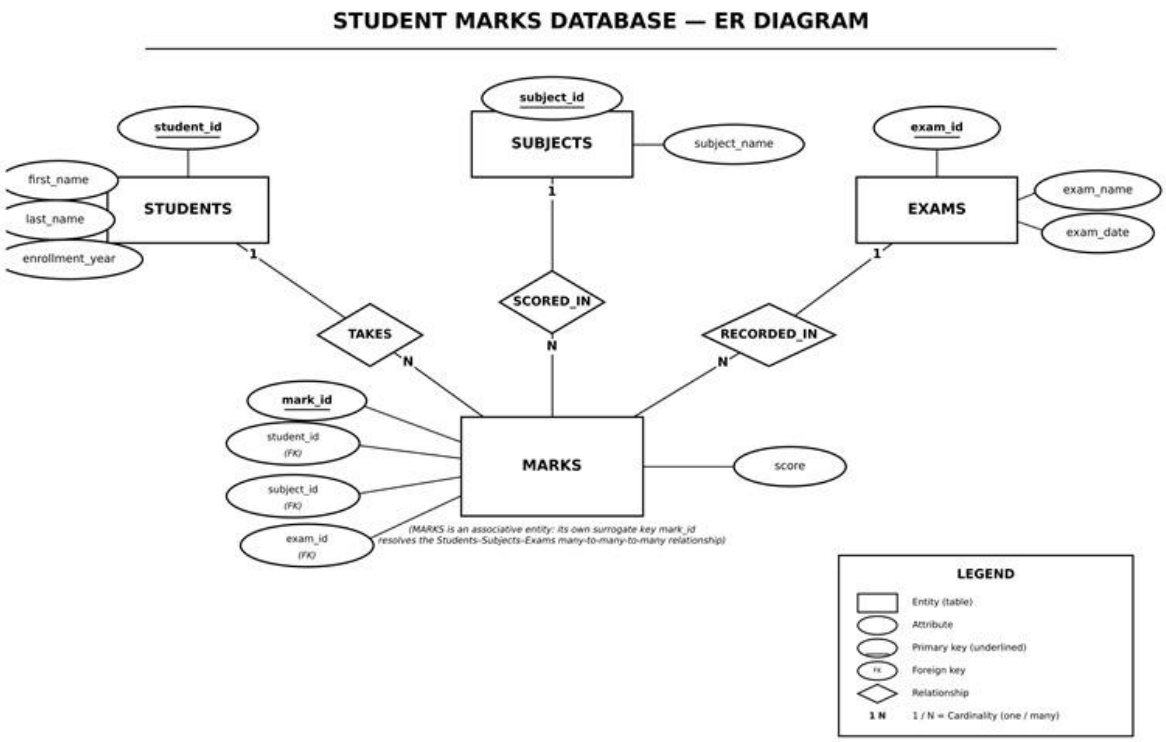
Database: studentdb | RDBMS: MySQL

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**1. Entity-Relationship (ER) Diagram**

STUDENTS, SUBJECTS and EXAMS are independent (strong) entities, each with its own primary key. MARKS is an associative entity: it carries its own surrogate key (mark_id) and three foreign keys, resolving the natural many-to-many-to-many relationship between a student, a subject and an exam.




### 1.1 Entity Summary

| Entity | Type | Primary Key | Notes |
|---|---|---|---|
| STUDENTS | Strong | student_id | Independent entity; a learner enrolled in a given year. |
| SUBJECTS | Strong | subject_id | Independent entity; a course/subject offered. |
| EXAMS | Strong | exam_id | Independent entity; a specific exam sitting with a date. |
| MARKS | Associative (strong, own PK) | mark_id | Foreign keys to all three: student_id, subject_id, exam_id. Represents one recorded score. |

### 2. Project Overview

STUDENT MARKS DATABASE | END-TO-END MYSQL PROJECT

RDBMS : MySQL 8.0+ (also compatible with MariaDB 10.x)

Notes : Every ANSWER below each query is a comment showing what

that query returns when run against the sample data in

**Contents:**


- PART 1 : Database & Table Creation (schema)
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- PART 10 : Query Bank - New Queries (Q10-Q21) + Answers
- PART 11 : Demo calls for Views / Procedures / Triggers
- PART 2. Run the SELECT yourself to confirm.


**DATABASE & TABLE CREATION**

```
DROP DATABASE IF EXISTS studentdb;
CREATE DATABASE studentdb;
USE studentdb;

-- ---------- Strong Entity: STUDENTS ----------
CREATE TABLE Students (
    student_id      INT PRIMARY KEY,
    first_name      VARCHAR(50),
    last_name       VARCHAR(50),
```

```
    enrollment_year INT
);

-- ---------- Strong Entity: SUBJECTS ----------
CREATE TABLE Subjects (
    subject_id   INT PRIMARY KEY,
    subject_name VARCHAR(50)
);

-- ---------- Strong Entity: EXAMS ----------
CREATE TABLE Exams (
    exam_id   INT PRIMARY KEY,
    exam_name VARCHAR(50),
    exam_date DATE
);

-- ---------- Associative Entity: MARKS ----------
-- Resolves the natural many-to-many-to-many relationship between
-- STUDENTS, SUBJECTS and EXAMS. It has its own surrogate key (mark_id),
-- so it is a strong/associative entity, not a weak entity -- but every
-- row is meaningless without its three foreign keys.
CREATE TABLE Marks (
    mark_id    INT PRIMARY KEY,
    student_id INT,
    subject_id INT,
    exam_id    INT,
    score      INT,
    FOREIGN KEY (student_id) REFERENCES Students(student_id),
```

```
    FOREIGN KEY (subject_id) REFERENCES Subjects(subject_id),
    FOREIGN KEY (exam_id)    REFERENCES Exams(exam_id)
);
```

**SAMPLE DATA**

```
-- Students
INSERT INTO Students VALUES
(1, 'Aarav',  'Sharma', 2023),
(2, 'Diya',   'Patel',  2023),
(3, 'Vihaan', 'Rao',    2024),
(4, 'Ananya', 'Singh',  2024);

-- Subjects
INSERT INTO Subjects VALUES
(101, 'Mathematics'),
(102, 'Science'),
(103, 'Hindi Literature');

-- Exams
INSERT INTO Exams VALUES
(1, 'Half-Yearly 2024',  '2024-09-15'),
(2, 'Final Annual 2024', '2024-03-20');

-- Marks
INSERT INTO Marks VALUES
(1, 1, 101, 1, 85),  -- Aarav,  Math,    Half-Yearly
```

```
(2, 1, 102, 1, 90),  -- Aarav,  Science, Half-Yearly
(3, 2, 101, 1, 75),  -- Diya,   Math,    Half-Yearly
(4, 2, 102, 1, 60),  -- Diya,   Science, Half-Yearly
(5, 3, 101, 1, 95),  -- Vihaan, Math,    Half-Yearly
(6, 1, 101, 2, 88),  -- Aarav,  Math,    Final
(7, 2, 101, 2, 79),  -- Diya,   Math,    Final
(8, 3, 101, 2, 92);  -- Vihaan, Math,    Final

-- Note: Ananya Singh (student_id 4) has no marks recorded, and
-- Hindi Literature (subject_id 103) is never used -- both are useful
-- edge cases for the LEFT JOIN / "never took an exam" queries below.
```

**INDEXES**

```
-- Foreign-key and frequently-filtered columns get explicit indexes to
-- speed up the JOINs, WHERE and GROUP BY clauses used throughout.

CREATE INDEX idx_marks_student     ON Marks(student_id);
CREATE INDEX idx_marks_subject     ON Marks(subject_id);
CREATE INDEX idx_marks_exam        ON Marks(exam_id);
CREATE INDEX idx_marks_score       ON Marks(score);
CREATE INDEX idx_students_enrollment ON Students(enrollment_year);
CREATE INDEX idx_exams_date        ON Exams(exam_date);

-- Verify indexes:
-- SHOW INDEX FROM Marks;
-- SHOW INDEX FROM Students;
```

**VIEWS**

```
-- 4.1 Flattened mark + student + subject + exam info (for reporting/UI)
CREATE OR REPLACE VIEW view_student_marks AS
SELECT
        M.mark_id,
        CONCAT(St.first_name, ' ', St.last_name) AS student_name,
        Sb.subject_name,
        E.exam_name,
        E.exam_date,
        M.score
FROM Marks M
JOIN Students St ON M.student_id = St.student_id
JOIN Subjects Sb ON M.subject_id = Sb.subject_id
JOIN Exams E   ON M.exam_id           = E.exam_id;

-- 4.2 Average score per subject
CREATE OR REPLACE VIEW view_subject_averages AS
SELECT
        Sb.subject_id,
        Sb.subject_name,
        ROUND(AVG(M.score), 2) AS average_score,
        COUNT(M.mark_id)    AS marks_recorded
FROM Subjects Sb
LEFT JOIN Marks M ON Sb.subject_id = M.subject_id
GROUP BY Sb.subject_id, Sb.subject_name;
```

```
-- 4.3 Per-student totals (includes students with zero marks)
CREATE OR REPLACE VIEW view_student_totals AS
SELECT
        St.student_id,
        CONCAT(St.first_name, ' ', St.last_name) AS student_name,
        St.enrollment_year,
        COUNT(M.mark_id)            AS marks_count,
        COALESCE(SUM(M.score), 0)           AS total_score,
        ROUND(AVG(M.score), 2)              AS average_score
FROM Students St
LEFT JOIN Marks M ON St.student_id = M.student_id
GROUP BY St.student_id, St.first_name, St.last_name, St.enrollment_year;

-- 4.4 Pass / Fail report (score >= 70 is a Pass)
CREATE OR REPLACE VIEW view_pass_fail AS
SELECT
        CONCAT(St.first_name, ' ', St.last_name) AS student_name,
        Sb.subject_name,
        E.exam_name,
        M.score,
        CASE WHEN M.score >= 70 THEN 'Pass' ELSE 'Fail' END AS status
FROM Marks M
JOIN Students St ON M.student_id = St.student_id
JOIN Subjects Sb ON M.subject_id = Sb.subject_id
JOIN Exams E   ON M.exam_id          = E.exam_id;

-- 4.5 Exam-wise summary (total score, pass count, fail count)
CREATE OR REPLACE VIEW view_exam_summary AS
```

```
SELECT
        E.exam_id,
        E.exam_name,
        SUM(M.score)                              AS total_score,
        SUM(CASE WHEN M.score >= 70 THEN 1 ELSE 0 END)      AS pass_count,
        SUM(CASE WHEN M.score < 70  THEN 1 ELSE 0 END)      AS fail_count
FROM Exams E
JOIN Marks M ON E.exam_id = M.exam_id
GROUP BY E.exam_id, E.exam_name;
```

**STORED PROCEDURES**

```
-- 5.1 All marks for one student (by student_id)
DROP PROCEDURE IF EXISTS sp_get_student_marks;
DELIMITER $$
CREATE PROCEDURE sp_get_student_marks(IN p_student_id INT)
BEGIN
        SELECT Sb.subject_name, E.exam_name, M.score
        FROM Marks M
        JOIN Subjects Sb ON M.subject_id = Sb.subject_id
        JOIN Exams E   ON M.exam_id          = E.exam_id
        WHERE M.student_id = p_student_id
        ORDER BY E.exam_date, Sb.subject_name;
END $$
DELIMITER ;

-- 5.2 Add a new mark (validates score range 0-100, auto-generates mark_id)
```

```
DROP PROCEDURE IF EXISTS sp_add_mark;
DELIMITER $$
CREATE PROCEDURE sp_add_mark(
        IN p_student_id INT,
        IN p_subject_id INT,
        IN p_exam_id  INT,
        IN p_score      INT
)
BEGIN
        DECLARE v_new_id INT;

        IF p_score < 0 OR p_score > 100 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Score must be between 0 and 100';
        ELSE
        SELECT COALESCE(MAX(mark_id), 0) + 1 INTO v_new_id FROM Marks;
        INSERT INTO Marks (mark_id, student_id, subject_id, exam_id, score)
        VALUES (v_new_id, p_student_id, p_subject_id, p_exam_id, p_score);
        END IF;
END $$
DELIMITER ;

-- 5.3 Average score for one subject (by subject_id)
DROP PROCEDURE IF EXISTS sp_subject_average;
DELIMITER $$
CREATE PROCEDURE sp_subject_average(IN p_subject_id INT)
BEGIN
        SELECT Sb.subject_name, ROUND(AVG(M.score), 2) AS average_score
```

```
        FROM Subjects Sb
        LEFT JOIN Marks M ON Sb.subject_id = M.subject_id
        WHERE Sb.subject_id = p_subject_id
        GROUP BY Sb.subject_name;
END $$
DELIMITER ;

-- 5.4 Full exam report: every mark + pass/fail + exam-level pass count
DROP PROCEDURE IF EXISTS sp_exam_report;
DELIMITER $$
CREATE PROCEDURE sp_exam_report(IN p_exam_id INT)
BEGIN
        SELECT CONCAT(St.first_name, ' ', St.last_name) AS student_name,
        Sb.subject_name,
        M.score,
        CASE WHEN M.score >= 70 THEN 'Pass' ELSE 'Fail' END AS status
        FROM Marks M
        JOIN Students St ON M.student_id = St.student_id
        JOIN Subjects Sb ON M.subject_id = Sb.subject_id
        WHERE M.exam_id = p_exam_id
        ORDER BY M.score DESC;
END $$
DELIMITER ;

-- 5.5 Student report card (all marks + overall average)
DROP PROCEDURE IF EXISTS sp_student_report_card;
DELIMITER $$
CREATE PROCEDURE sp_student_report_card(IN p_student_id INT)
```

```
BEGIN
        SELECT Sb.subject_name, E.exam_name, M.score,
        CASE WHEN M.score >= 70 THEN 'Pass' ELSE 'Fail' END AS status
        FROM Marks M
        JOIN Subjects Sb ON M.subject_id = Sb.subject_id
        JOIN Exams E   ON M.exam_id          = E.exam_id
        WHERE M.student_id = p_student_id;

        SELECT ROUND(AVG(score), 2) AS overall_average
        FROM Marks
        WHERE student_id = p_student_id;
END $$
DELIMITER ;
```

**TRIGGERS**

```
-- 6.0 Supporting audit table for mark changes
CREATE TABLE marks_log (
        log_id      INT AUTO_INCREMENT PRIMARY KEY,
        mark_id     INT,
        student_id  INT,
        action_type VARCHAR(10),
        old_score   INT,
        new_score   INT,
        change_date DATETIME DEFAULT CURRENT_TIMESTAMP
);
```

```
-- 6.1 Reject an out-of-range score before it is ever inserted
DROP TRIGGER IF EXISTS trg_validate_score_insert;
DELIMITER $$
CREATE TRIGGER trg_validate_score_insert
BEFORE INSERT ON Marks
FOR EACH ROW
BEGIN
        IF NEW.score < 0 OR NEW.score > 100 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Score must be between 0 and 100';
        END IF;
END $$
DELIMITER ;

-- 6.2 Reject an out-of-range score on UPDATE too
DROP TRIGGER IF EXISTS trg_validate_score_update;
DELIMITER $$
CREATE TRIGGER trg_validate_score_update
BEFORE UPDATE ON Marks
FOR EACH ROW
BEGIN
        IF NEW.score < 0 OR NEW.score > 100 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'Score must be between 0 and 100';
        END IF;
END $$
DELIMITER ;
```

```
-- 6.3 Block duplicate marks (same student + subject + exam already exists)
DROP TRIGGER IF EXISTS trg_prevent_duplicate_mark;
DELIMITER $$
CREATE TRIGGER trg_prevent_duplicate_mark
BEFORE INSERT ON Marks
FOR EACH ROW
BEGIN
        DECLARE v_count INT;
        SELECT COUNT(*) INTO v_count
        FROM Marks
        WHERE student_id = NEW.student_id
        AND subject_id = NEW.subject_id
        AND exam_id  = NEW.exam_id;

        IF v_count > 0 THEN
        SIGNAL SQLSTATE '45000'
        SET MESSAGE_TEXT = 'This student already has a mark for this subject in this exam';
        END IF;
END $$
DELIMITER ;

-- 6.4 Log every new mark inserted
DROP TRIGGER IF EXISTS trg_log_mark_insert;
DELIMITER $$
CREATE TRIGGER trg_log_mark_insert
AFTER INSERT ON Marks
FOR EACH ROW
BEGIN
```

```
    INSERT INTO marks_log (mark_id, student_id, action_type, old_score, new_score)
    VALUES (NEW.mark_id, NEW.student_id, 'INSERT', NULL, NEW.score);
END $$
DELIMITER ;

-- 6.5 Log every score change (old vs new)
DROP TRIGGER IF EXISTS trg_log_mark_update;
DELIMITER $$
CREATE TRIGGER trg_log_mark_update
AFTER UPDATE ON Marks
FOR EACH ROW
BEGIN
    IF OLD.score <> NEW.score THEN
    INSERT INTO marks_log (mark_id, student_id, action_type, old_score, new_score)
    VALUES (NEW.mark_id, NEW.student_id, 'UPDATE', OLD.score, NEW.score);
    END IF;
END $$
DELIMITER ;

-- 6.6 Log every deleted mark
DROP TRIGGER IF EXISTS trg_log_mark_delete;
DELIMITER $$
CREATE TRIGGER trg_log_mark_delete
AFTER DELETE ON Marks
FOR EACH ROW
BEGIN
    INSERT INTO marks_log (mark_id, student_id, action_type, old_score, new_score)
    VALUES (OLD.mark_id, OLD.student_id, 'DELETE', OLD.score, NULL);
```

```
END $$
DELIMITER ;
```

**QUERY BANK - LEVEL 1 : SIMPLE QUERIES (Original Q1-Q3)**

```
-- Q1. Retrieve all students (first and last name).
SELECT first_name, last_name FROM Students;
-- ANSWER:
--   Aarav Sharma, Diya Patel, Vihaan Rao, Ananya Singh

-- Q2. Filter by Score: marks greater than 80.
SELECT * FROM Marks WHERE score > 80;
-- ANSWER:
--   mark_id 1 - Aarav  - Math    - Half-Yearly - 85
--   mark_id 2 - Aarav  - Science - Half-Yearly - 90
--   mark_id 5 - Vihaan - Math - Half-Yearly - 95
--   mark_id 6 - Aarav  - Math    - Final          - 88
--   mark_id 8 - Vihaan - Math - Final            - 92

-- Q3. Sort Data: all exams, most recent first.
SELECT * FROM Exams ORDER BY exam_date DESC;
-- ANSWER:
--   Half-Yearly 2024  - 2024-09-15
--   Final Annual 2024 - 2024-03-20
```

**QUERY BANK - LEVEL 2 : INTERMEDIATE QUERIES (Original Q4-Q6)**

```
-- Q4. Student Marks: First Name, Subject Name, Score for every mark.
SELECT St.first_name, Sb.subject_name, M.score
FROM Marks M
JOIN Students St ON M.student_id = St.student_id
JOIN Subjects Sb ON M.subject_id = Sb.subject_id;
-- ANSWER: 8 rows -- Aarav/Math/85, Aarav/Science/90, Diya/Math/75,
--   Diya/Science/60, Vihaan/Math/95, Aarav/Math/88, Diya/Math/79, Vihaan/Math/92

-- Q5. Subject Averages: average score per subject.
SELECT Sb.subject_name, AVG(M.score) AS average_score
FROM Marks M
JOIN Subjects Sb ON M.subject_id = Sb.subject_id
GROUP BY Sb.subject_name;
-- ANSWER:
--   Mathematics - 85.67  (85+75+95+88+79+92 = 514 / 6)
--   Science     - 75.00  (90+60 = 150 / 2)
--   (Hindi Literature does not appear -- an INNER JOIN excludes subjects
--         with zero marks recorded; see view_subject_averages in PART 4 for
--         a version that includes it via LEFT JOIN.)

-- Q6. Count Exams: how many marks recorded for "Aarav Sharma"?
SELECT COUNT(*) AS total_exams_taken
FROM Marks M
JOIN Students St ON M.student_id = St.student_id
WHERE St.first_name = 'Aarav' AND St.last_name = 'Sharma';
-- ANSWER: 3  (Math/Half-Yearly, Science/Half-Yearly, Math/Final)
```

**QUERY BANK - LEVEL 3 : COMPLEX QUERIES (Original Q7-Q9)**

```
-- Q7. Top Performer: student with the highest single score in any exam.
SELECT first_name, last_name
FROM Students
WHERE student_id = (
        SELECT student_id FROM Marks ORDER BY score DESC LIMIT 1
);
-- Note: use 'TOP 1' instead of 'LIMIT 1' if you are on MS SQL Server.
-- ANSWER: Vihaan Rao (score 95, Mathematics, Half-Yearly 2024)

-- Q8. Scores by Exam: exam name + total score, only where total > 250.
SELECT E.exam_name, SUM(M.score) AS total_score
FROM Marks M
JOIN Exams E ON M.exam_id = E.exam_id
GROUP BY E.exam_name
HAVING SUM(M.score) > 250;
-- ANSWER:
--   Half-Yearly 2024  - 405  (85+90+75+60+95)
--   Final Annual 2024 - 259  (88+79+92)
--   (Both exams clear the 250 threshold.)

-- Q9. Pass/Fail Label: Student Name, Subject, Score, Pass/Fail status.
SELECT
        St.first_name, Sb.subject_name, M.score,
        CASE WHEN M.score >= 70 THEN 'Pass' ELSE 'Fail' END AS status
FROM Marks M
```

```
JOIN Students St ON M.student_id = St.student_id
JOIN Subjects Sb ON M.subject_id = Sb.subject_id;
-- ANSWER: 8 rows, all "Pass" except Diya / Science / 60 -> "Fail"
```

**QUERY BANK - NEW QUERIES (Q10-Q21)**

```
-- Q10. Students enrolled in 2023.
SELECT first_name, last_name FROM Students WHERE enrollment_year = 2023;
-- ANSWER: Aarav Sharma, Diya Patel

-- Q11. Highest score recorded in each subject.
SELECT Sb.subject_name, MAX(M.score) AS highest_score
FROM Marks M
JOIN Subjects Sb ON M.subject_id = Sb.subject_id
GROUP BY Sb.subject_name;
-- ANSWER:
--   Mathematics - 95  (Vihaan Rao, Half-Yearly)
--   Science     - 90  (Aarav Sharma, Half-Yearly)

-- Q12. Students who took the 'Final Annual 2024' exam.
SELECT DISTINCT St.first_name, St.last_name
FROM Marks M
JOIN Students St ON M.student_id = St.student_id
JOIN Exams E ON M.exam_id = E.exam_id
WHERE E.exam_name = 'Final Annual 2024';
-- ANSWER: Aarav Sharma, Diya Patel, Vihaan Rao  (Ananya Singh did not)
```

```sql
-- Q13. Students whose average score (across all their marks) is above 80.
SELECT St.first_name, St.last_name, ROUND(AVG(M.score), 2) AS avg_score
FROM Marks M
JOIN Students St ON M.student_id = St.student_id
GROUP BY St.student_id, St.first_name, St.last_name
HAVING AVG(M.score) > 80;
-- ANSWER:
--   Vihaan Rao   - 93.50  ( (95+92)/2 )
--   Aarav Sharma - 87.67  ( (85+90+88)/3 )

-- Q14. Number of students per enrollment year.
SELECT enrollment_year, COUNT(*) AS student_count
FROM Students
GROUP BY enrollment_year;
-- ANSWER:
--   2023 - 2 students  (Aarav, Diya)
--   2024 - 2 students  (Vihaan, Ananya)

-- Q15. Which subject has the highest average score?
SELECT Sb.subject_name, ROUND(AVG(M.score), 2) AS avg_score
FROM Marks M
JOIN Subjects Sb ON M.subject_id = Sb.subject_id
GROUP BY Sb.subject_name
ORDER BY avg_score DESC
LIMIT 1;
-- ANSWER: Mathematics - 85.67

-- Q16. Students who scored above their subject's own average (correlated subquery).
```

```
SELECT St.first_name, Sb.subject_name, M.score
FROM Marks M
JOIN Students St ON M.student_id = St.student_id
JOIN Subjects Sb ON M.subject_id = Sb.subject_id
WHERE M.score > (
        SELECT AVG(M2.score) FROM Marks M2 WHERE M2.subject_id = M.subject_id
);
-- ANSWER:
--   Aarav  - Science - 90  (subject avg 75.00)
--   Vihaan - Math    - 95  (subject avg 85.67)
--   Aarav  - Math    - 88  (subject avg 85.67)
--   Vihaan - Math    - 92  (subject avg 85.67)

-- Q17. Pass count per exam (score >= 70).
SELECT E.exam_name,
        SUM(CASE WHEN M.score >= 70 THEN 1 ELSE 0 END) AS pass_count,
        SUM(CASE WHEN M.score < 70  THEN 1 ELSE 0 END) AS fail_count
FROM Marks M
JOIN Exams E ON M.exam_id = E.exam_id
GROUP BY E.exam_name;
-- ANSWER:
--   Half-Yearly 2024  - 4 pass, 1 fail  (fail: Diya/Science/60)
--   Final Annual 2024 - 3 pass, 0 fail

-- Q18. Full detail list: student, subject, exam, score (3-way join).
SELECT CONCAT(St.first_name, ' ', St.last_name) AS student_name,
        Sb.subject_name, E.exam_name, M.score
FROM Marks M
```

```
JOIN Students St ON M.student_id = St.student_id
JOIN Subjects Sb ON M.subject_id = Sb.subject_id
JOIN Exams E   ON M.exam_id           = E.exam_id
ORDER BY E.exam_date, St.first_name;
-- ANSWER: all 8 marks fully detailed (same data as view_student_marks)

-- Q19. Students who have never taken any exam (no marks recorded).
SELECT St.first_name, St.last_name
FROM Students St
LEFT JOIN Marks M ON St.student_id = M.student_id
WHERE M.mark_id IS NULL;
-- ANSWER: Ananya Singh (the only student with zero marks)

-- Q20. Rank all students by total score, highest first.
SELECT CONCAT(St.first_name, ' ', St.last_name) AS student_name,
          COALESCE(SUM(M.score), 0) AS total_score
FROM Students St
LEFT JOIN Marks M ON St.student_id = M.student_id
GROUP BY St.student_id, St.first_name, St.last_name
ORDER BY total_score DESC;
-- ANSWER:
--   1. Aarav Sharma - 263  (85+90+88)
--   2. Diya Patel   - 214  (75+60+79)
--   3. Vihaan Rao   - 187  (95+92)
--   4. Ananya Singh - 0(no marks recorded)

-- Q21. Second-highest individual score recorded.
SELECT DISTINCT score
```

```
FROM Marks

ORDER BY score DESC

LIMIT 1 OFFSET 1;

-- ANSWER: 92  (highest is 95; second-highest is 92, Vihaan Rao, Final Math)
```

**DEMO CALLS - Views / Stored Procedures / Triggers**

```
-- Run these one at a time to see the objects from PART 4-6 in action.

-- Views:

-- SELECT * FROM view_student_marks;

-- SELECT * FROM view_subject_averages;

-- SELECT * FROM view_student_totals ORDER BY total_score DESC;

-- SELECT * FROM view_pass_fail WHERE status = 'Fail';

-- SELECT * FROM view_exam_summary;

-- Stored Procedures:

-- CALL sp_get_student_marks(1);              -- Aarav Sharma's marks

-- CALL sp_add_mark(4, 103, 1, 72);           -- Ananya gets a Hindi Literature mark

-- CALL sp_subject_average(101);              -- Mathematics average

-- CALL sp_exam_report(2);                    -- Final Annual 2024 report

-- CALL sp_student_report_card(2);            -- Diya Patel's report card

-- Triggers (fire automatically, nothing to CALL):

--   * Inserting a Mark with score < 0 or > 100 is blocked      -> trg_validate_score_insert

--   * Updating a Mark to an out-of-range score is blocked    -> trg_validate_score_update

--   * Inserting a duplicate (same student+subject+exam) is

--         blocked                                  -> trg_prevent_duplicate_mark
```

```
--   * Every INSERT / UPDATE / DELETE on Marks is written to
--         marks_log automatically                    -> trg_log_mark_insert /
--                                                    trg_log_mark_update /
--                                                    trg_log_mark_delete


-- Check the audit trail after running the demo calls above:
-- SELECT * FROM marks_log ORDER BY log_id;


-- ================================================================
-- END OF PROJECT SCRIPT
-- ================================================================
```